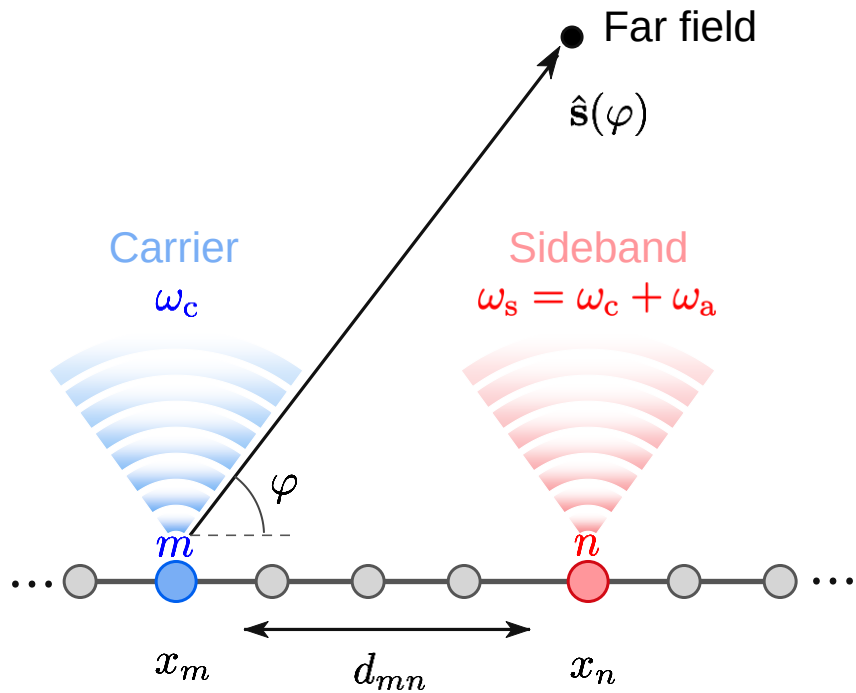


a Multi-channel interaction



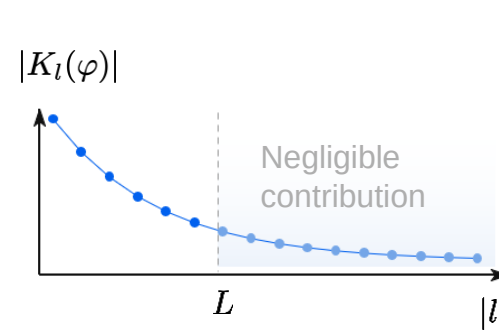
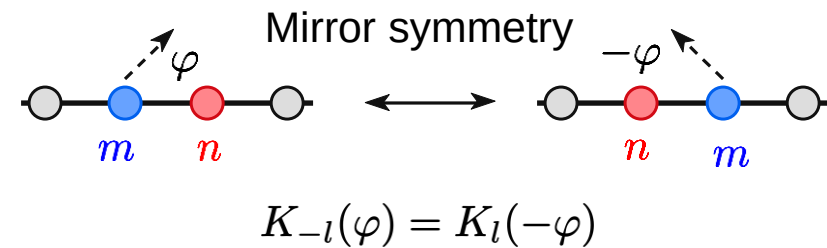
b Structure of the pair response

$$H_{mn}(\varphi) = \underbrace{e^{-ik_a x_m \sin \varphi}}_{\text{Absolute position phase}} \cdot \underbrace{K_{n-m}(\varphi)}_{\text{Relative displacement kernel}}$$

$$\mathbf{T}(\varphi) = \begin{bmatrix} K_0(\varphi) & K_1(\varphi) & K_2(\varphi) & \cdots & K_{N-1}(\varphi) \\ K_{-1}(\varphi) & K_0(\varphi) & K_1(\varphi) & \cdots & K_{N-2}(\varphi) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ K_{-(N-1)}(\varphi) & K_{-(N-2)}(\varphi) & K_{-(N-3)}(\varphi) & \cdots & K_0(\varphi) \end{bmatrix}$$

$$\mathbf{H}(\varphi) = \Phi(\varphi) \mathbf{T}(\varphi)$$

c Symmetry and banded approximation



Interaction decay

$$\mathbf{T}(\varphi) \approx \begin{bmatrix} & & & 0 \\ & & & \\ & & & \\ 0 & & & \end{bmatrix} \quad |n - m| \leq L$$

Banded truncation

$$N^2 \xrightarrow{\text{Translation invariance}} 2N-1 \xrightarrow{\text{Mirror symmetry}} N \xrightarrow{\text{Banded truncation}} L+1$$